

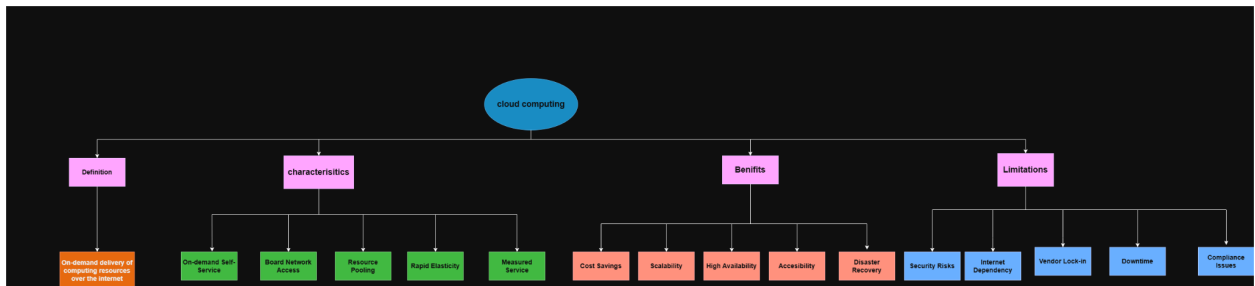
CSA1509-Cloud Computing

ASSESSMENT TOOL 1

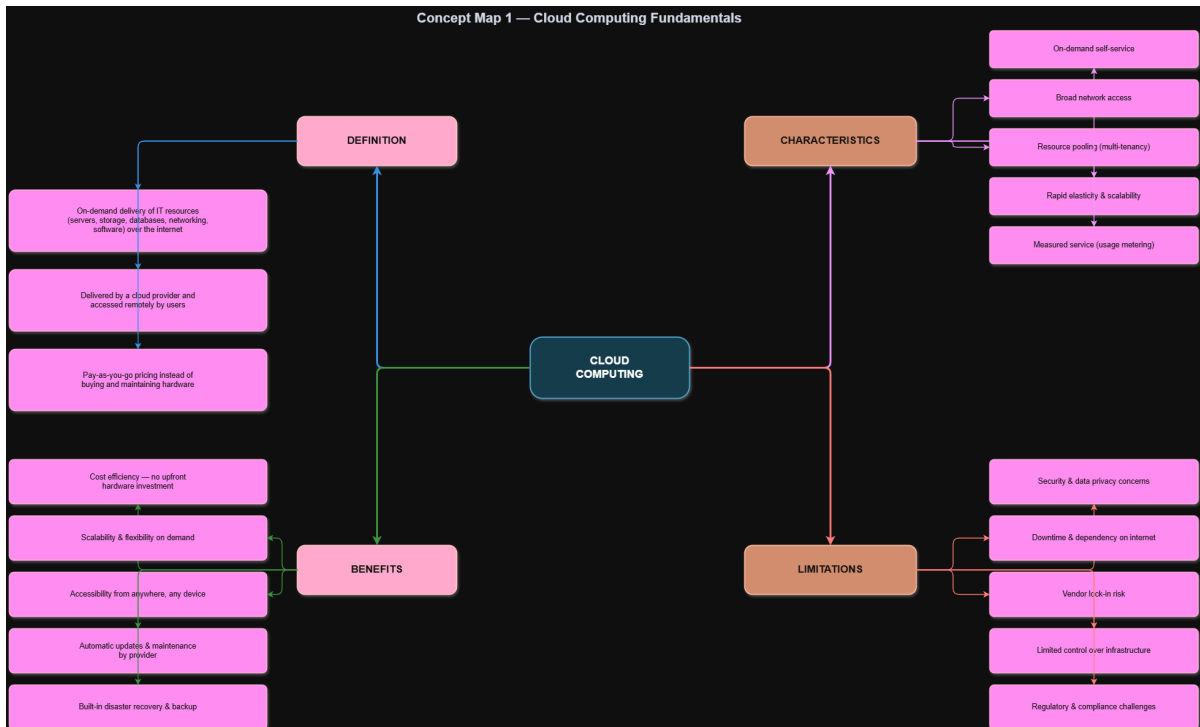
Assessment Tasks

Concept Mapping Tasks

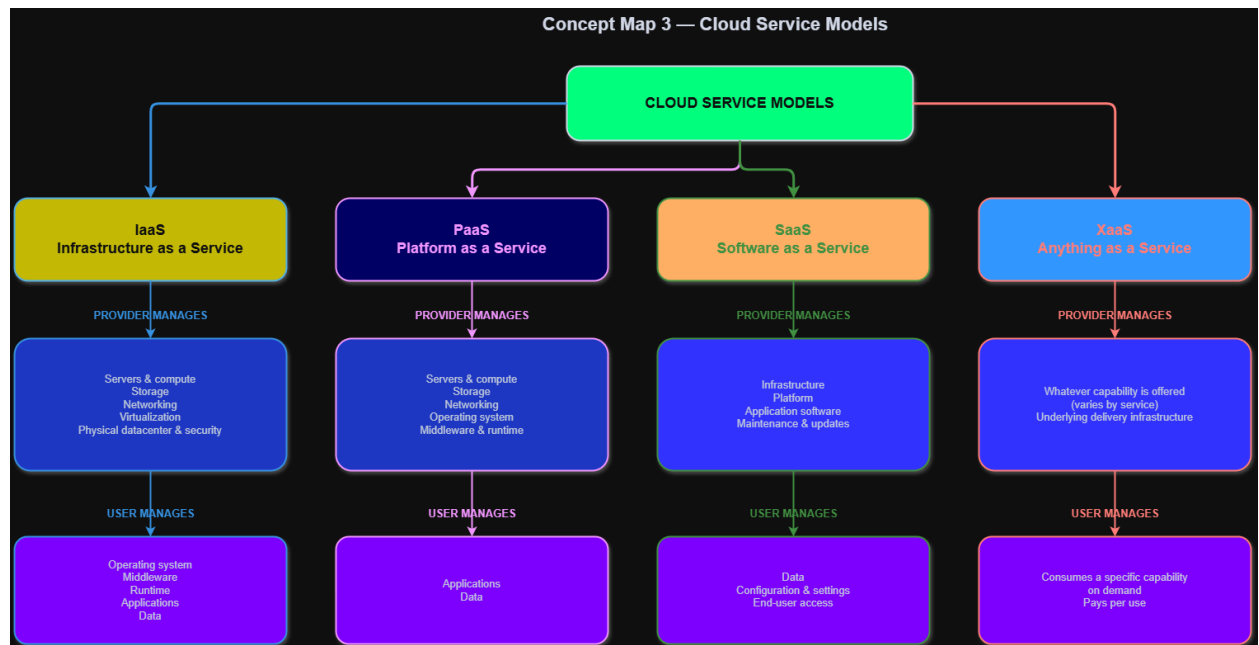
1. **Concept Map 1: Cloud Computing Fundamentals.** Include definition, characteristics, benefits, and limitations.



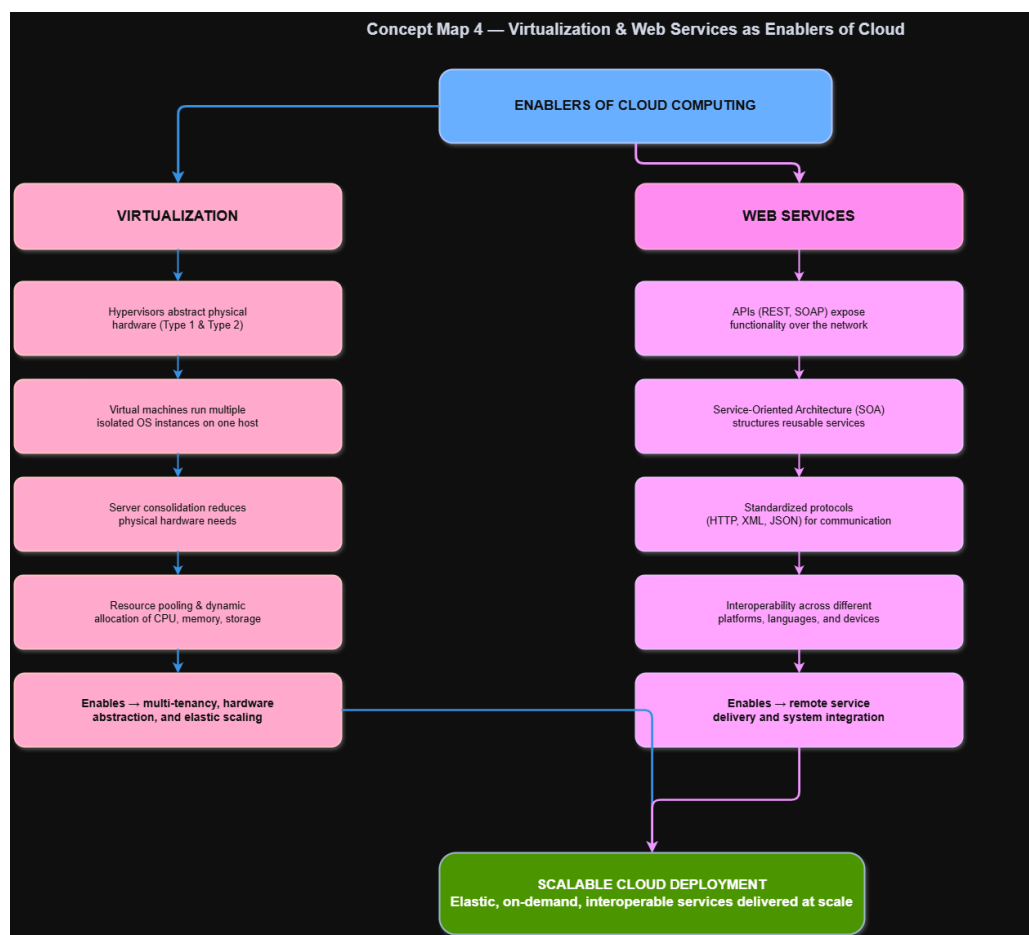
2. **Concept Map 2: Evolution of Cloud Computing.** Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. **Concept Map 3: Service Models.** Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. **Concept Map 4: Virtualization and Web Services as Enablers of Cloud.** Show how these two concepts support cloud deployment and scalability.



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents cloud computing by organizing the main concepts into clear categories such as cloud characteristics, service models, deployment models, and benefits. The use of connected branches makes it easy to understand the relationship between different cloud components. Design decisions such as color coding, hierarchical structure, and labeled connections improve readability and help explain how cloud computing works. These decisions make the concept map simple, logical, and easy to learn.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows the cloud system to handle increasing workloads by adding resources when required. Elasticity automatically increases or decreases resources based on demand, reducing unnecessary costs. Resource management ensures that computing resources are allocated efficiently among users. Without these capabilities, the system may experience slow performance, higher operational costs, resource wastage, and difficulty handling sudden increases in user traffic, leading to poor service quality.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes APIs, web services, and cloud services because they enable communication between applications and cloud resources. APIs allow software to access cloud functions easily, while web services support data exchange between different platforms. Cloud services such as IaaS, PaaS, and SaaS provide infrastructure, development platforms, and software applications without requiring local installation. These services improve flexibility, reduce maintenance, increase efficiency, and support faster application development.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The components in the concept map are interconnected to show how cloud services depend on networking, virtualization, storage, security, and resource management. Virtualization enables efficient hardware utilization, while security protects user data. Storage services ensure data availability, and networking provides reliable communication. Together, these interactions improve system reliability, increase application performance, reduce downtime, and provide users with faster, secure, and uninterrupted access to cloud services.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map improved my understanding of cloud computing by showing how different technologies and services work together in a single architecture. It helped me understand the relationships between service models, deployment models, virtualization, scalability, and security. If I were redesigning the concept map, I would include more real-world examples, detailed security mechanisms, cloud provider examples such as AWS, Microsoft Azure, and Google Cloud, and additional diagrams to make the architecture more informative and visually appealing.